NALDEBAIN (dinalbuphine sebacate) Extended Release Injection 75 mg/ mL.

FULL PRESCRIBING INFORMATION

To be Prescribed by Surgeon and/ or Anesthesiologist only.
For Hospital Use only.

1 INDICATIONS AND USAGE

NALDEBAIN is indicated for the relief of moderate to severe acute postsurgical pain.

2 DOSAGE AND ADMINISTRATION

General Dosing Information

Administer intramuscularly at a single dose of 150 mg.

It is not necessary to adjust dosage based on body surface area or body weight.

NALDEBAIN is an extended release formulation, it should be taken into consideration that it takes 12-24 hours to achieve therapeutic concentration.

NALDEBAIN is not adequate for administration in patients with urgent analgesics need.

NALDEBAIN is in a fixed dose package and is only for single dose use.

Safety and effectiveness for repeated-dose use have not been established.

The studies of concomitant use with other drugs including general anesthetic have not been conducted except for nalbuphine and ketorolac.

Instructions for Use

NALDEBAIN administered only via intramuscular route.

Instructions for administration:

1. Clean the vial top with an alcohol swab before use.
2. Draw up 2 mL of drug into syringe.
3. After preparing the skin, inject in the upper outer quadrant of the gluteus maximus. The solution is viscous and oily. Slow injection is recommended.
4. Slightly applying pressure to the injection site to prevent drug solution leakage.
5. Do not massage the injection site.

3 DOSAGE FORMS AND STRENGTHS

NALDEBAIN® ER Injection, 2 mL/vial is a sterile, clear and light yellow oily solution containing 75 mg/mL dinalbuphine sebacate. The product is supplied in a glass vial.

4 CONTRAINDICATION

NALDEBAIN administered via intramuscular route. It is prohibited for intravenous administration.

NALDEBAIN should not be administered to patients who are hypersensitive to nalbuphine, sesame oil or benzyl benzoate in NALDEBAIN.

5 WARNINGS AND PRECAUTION

Use in Ambulatory Patients

Nalbuphine hydrochloride may impair the mental or physical abilities required for the performance of potentially dangerous tasks such as driving a car or operating machinery. Therefore, NALDEBAIN should be administered with caution to ambulatory patients who should be warned to avoid such hazards.

Use in Emergency Procedures

Maintain patient under observation until recovered from nalbuphine hydrochloride effects that would affect driving or other potentially dangerous tasks.

Use in Pregnancy (Other Than Labor)

Severe fetal bradycardia has been reported when nalbuphine is administered during labor. Although there are no reports of developmental toxicity, including teratogenicity, or harm to the fetus in reproduction studies, this drug should be used in pregnancy only if clearly needed, if the potential benefit outweighs the risk to the fetus.

Use During Labor and Delivery

The placental transfer of nalbuphine is high, rapid, and variable with a maternal to fetal ratio ranging from 1:0.37 to 1:6. Fetal and neonatal adverse effects that have been reported following the administration of nalbuphine to the mother during labor include fetal bradycardia, respiratory depression at birth, apnea, cyanosis, and hypotonia. Some of these events have been life-threatening. Maternal administration of naloxone during labor has normalized these effects in some cases. Severe and prolonged fetal bradycardia has been reported. Permanent neurological damage attributed to fetal bradycardia has occurred. A sinusoidal fetal heart rate pattern associated with the use of nalbuphine has also been reported. Nalbuphine hydrochloride or NALDEBAIN should be used during labor and delivery only if clearly indicated and only if the potential benefit outweighs the risk to the infant. Newborns should be monitored for respiratory depression, apnea, bradycardia and arrhythmias if Nalbuphine hydrochloride or NALDEBAIN has been used.

Head Injury and Increased Intracranial Pressure

The possible respiratory depressant effects and the potential of potent analgesics to elevate cerebrospinal fluid pressure (resulting from vasodilation following CO_2 retention) may be markedly exaggerated in the presence of head injury, intracranial lesions or a preexisting increase in intracranial pressure. Furthermore, potent analgesics can produce effects which may obscure the clinical course of patients with head injuries. Therefore nalbuphine /NALDEBAIN should be used in these circumstances only when essential, and then should be administered with extreme caution.

Renal Impairment

NALDEBAIN does not need to adjust dosage in patients with renal impairment.

Hepatic Impairment

NALDEBAIN should be used with caution in patients with liver dysfunction. Because nalbuphine is metabolized in the liver and excreted by the kidneys, NALDEBAIN should be used with caution in patients with liver dysfunction.

Habit-forming

This preparation may be habit-forming on prolonged repeated use.

Risks from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Naldebainwith benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Naldebainis used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Naldebainwith serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue Naldebainif serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)

6 ADVERSE REACTIONS

Because clinical studies are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical studies of another drug and may not reflect the rates observed in practice.

Commonly-Observed Adverse Reactions in Double-Blind, Placebo-Controlled Clinical Trials

A total of 109 subjects received single dose of 150mg NALDEBAIN were included in the population for safety evaluation of NALDEBAIN.

Overall evaluation of the safety profile in clinical studies, the most clinically significant adverse reactions observed with NALDEBAIN 150mg were nausea, vomiting, injection site reaction, pyrexia and dizziness. All those reactions are assessed as mild to moderate in severity. The incidence of adverse reactions listed in Table 1.

Table 1 Summary of ADR Incidence

Adverse drug reaction	NALDEBAIN N=109		Placebo N=112	
	n	%	n	%
Injection site reaction	30	27.5%	7	6.3%
Pyrexia	18	16.5%	10	8.9%
Dizziness	7	6.4%	1	0.9%
Vomiting	3	2.8%	0	0.0%
Nausea	2	1.8%	0	0.0%
Somnolence	1	0.9%	0	0.0%

Most of the subjects in phase III studies with injection site reaction were recovered present at final visit (Day 7~10). In a bioavailability study, the study period after NALDEBAIN administration is 14 days. The injection site reaction is monitored until end of the study. Part of subjects recovered on Day 8, and all subjects recovered on Day 12. All the subjects felt the symptoms is tolerable and finished the study. The observation is all the injection site reaction recovered at the end of the study. That means the reaction is tolerable and reversible

Overall evaluation of the safety profile in clinical studies, the frequency of adverse events whether drug related or not in NALDEBAIN and placebo treatment groups is summarized in Table 2.

Table 2 Incidence of subjects with TEAE by body system-Safety population

System Organ Class Preferred Term	NALDEBAIN (N = 109)	Placebo (N = 112)	Overall (N = 221)
General disorders and administration site conditions			
Chills	1 (0.9%)	1 (0.9%)	2 (0.9%)
Fatigue	1 (0.9%)	3 (2.7%)	4 (1.8%)
Feeling cold	2 (1.8%)	0 (0.0%)	2 (0.9%)
Injection site swelling	0 (0.0%)	1 (0.9%)	1 (0.5%)
Pyrexia	41 (37.6%)	20 (17.9%)	61 (27.6%)
Gastrointestinal disorders			
Abdominal discomfort	0 (0.0%)	1 (0.9%)	1 (0.5%)
Abdominal distension	4 (3.7%)	3 (2.7%)	7 (3.2%)
Abdominal pain	1 (0.9%)	0 (0.0%)	1 (0.5%)
Abdominal pain lower	1 (0.9%)	0 (0.0%)	1 (0.5%)
Abdominal pain upper	2 (1.8%)	2 (1.8%)	4 (1.8%)
Anal pruritus	0 (0.0%)	1 (0.9%)	1 (0.5%)
Constipation	13 (11.9%)	12 (10.7%)	25 (11.3%)
System Organ Class Preferred Term	**NALDEBAIN (N = 109)**	**Placebo (N = 112)**	**Overall (N = 221)**
Diarrhoea	0 (0.0%)	1 (0.9%)	1 (0.5%)
Faecaloma	1 (0.9%)	0 (0.0%)	1 (0.5%)
Flatulence	1 (0.9%)	0 (0.0%)	1 (0.5%)
Gastrointestinal motility disorder	0 (0.0%)	1 (0.9%)	1 (0.5%)
Intestinal obstruction	1 (0.9%)	0 (0.0%)	1 (0.5%)
Irritable bowel syndrome	1 (0.9%)	7 (6.3%)	8 (3.6%)
Nausea	5 (4.6%)	3 (2.7%)	8 (3.6%)
Oesophageal ulcer	1 (0.9%)	0 (0.0%)	1 (0.5%)
Vomiting	10 (9.2%)	1 (0.9%)	11 (5.0%)
Renal and urinary disorders			
Cystitis noninfective	1 (0.9%)	0 (0.0%)	1 (0.5%)
Dysuria	12 (11.0%)	11 (9.8%)	23 (10.4%)
Urinary retention	6 (5.5%)	6 (5.4%)	12 (5.4%)
Nervous system disorders			
Dizziness	18 (16.5%)	4 (3.6%)	22 (10.0%)
Headache	4 (3.7%)	4 (3.6%)	8 (3.6%)
Hypoaesthesia	1 (0.9%)	0 (0.0%)	1 (0.5%)
Poor quality sleep	0 (0.0%)	2 (1.8%)	2 (0.9%)
Somnolence	1 (0.9%)	0 (0.0%)	1 (0.5%)
Psychiatric disorders			
Anxiety	1 (0.9%)	5 (4.5%)	6 (2.7%)
Insomnia	1 (0.9%)	5 (4.5%)	6 (2.7%)
Nervousness	1 (0.9%)	0 (0.0%)	1 (0.5%)
Injury, poisoning and procedural complications			
Post procedural haemorrhage	1 (0.9%)	2 (1.8%)	3 (1.4%)
Post procedural swelling	3 (2.8%)	1 (0.9%)	4 (1.8%)
Infections and infestations			
Urinary tract infection	1 (0.9%)	0 (0.0%)	1 (0.5%)
Nasopharyngitis	0 (0.0%)	1 (0.9%)	1 (0.5%)
Urinary tract infection	1 (0.9%)	2 (1.8%)	3 (1.4%)
Investigations			
Blood pressure decreased	2 (1.8%)	0 (0%)	2 (0.9%)
Blood pressure systolic decreased	1 (0.9%)	0 (0%)	1 (0.5%)
Liver function test abnormal	2 (1.8%)	0 (0%)	2 (0.9%)
Respiratory, thoracic and mediastinal disorders			
Cough	2 (1.8%)	2 (1.8%)	4 (1.8%)
Rhinorrhoea	0 (0.0%)	1 (0.9%)	1 (0.5%)
Skin and subcutaneous tissue disorders			
Eczema	0 (0%)	1 (0.9%)	1 (0.5%)
Hyperhidrosis	1 (0.9%)	0 (0%)	1 (0.5%)
Rash pruritic	0 (0%)	1 (0.9%)	1 (0.5%)
Urticaria	1 (0.9%)	0 (0%)	1 (0.5%)
Ear and labyrinth disorders			
Vertigo	2 (1.8%)	0 (0%)	2 (0.9%)
Eye disorders			
Conjunctival pallor	0 (0%)	1 (0.9%)	1 (0.5%)
Scleral haemorrhage	0 (0%)	1 (0.9%)	1 (0.5%)
Blood and lymphatic system disorders			
Anaemia	0 (0%)	1 (0.9%)	1 (0.5%)
Cardiac disorders			
Palpitations	1 (0.5%)	0 (0%)	1 (0.5%)
Metabolism and nutrition disorders			
Decreased appetite	0 (0%)	1 (0.9%)	1 (0.5%)
Musculoskeletal and connective tissue discorders			
System Organ Class Preferred Term	**NALDEBAIN (N = 109)**	**Placebo (N = 112)**	**Overall (N = 221)**
Myalgia	0 (0%)	1 (0.9%)	1 (0.5%)
Vascular disorders			
Hypertension	0 (0%)	1 (0.9%)	1 (0.5%)

Postmarketing Experience

Serotonin syndrome (See Warnings and Precautions) Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamicpituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

7 DRUG INTERACTIONS

Central Nervous System Depressants

Studies of NALDEBAIN concomitant use with general anesthetic have not been conducted.

Although nalbuphine possesses opioid antagonist activity, there is evidence that in nondependent patients it will not antagonize an opioid analgesic administered just before, concurrently, or just after an injection of nalbuphine. Therefore, patients receiving an opioid analgesic, general anesthetics, phenothiazines, or other tranquilizers, sedatives, hypnotics, or other CNS depressants (including alcohol) concomitantly with NALDEBAIN may exhibit an additive effect. When such combined therapy is contemplated, the dose of one or both agents should be reduced.

In phase III studies, all subjects were given local anesthetic (bupivacaine) prior to surgery(99%). 94% of subjects combined using local anesthetics Lidocaine. About 2 % subjects used midazolam. Reviewing overall adverse events, administration of local general anesthetics, bupivacaine, lidocaine, propofol and midazolam in combination with NALDEBAIN in phase III studies does not result in clinical significant adverse reactions.

Opioids

Studies of NALDEBAIN concomitant use with opiates have not been conducted. Since NALDEBAIN is nalbuphine's prodrug, the concomitant use with opioids could refer to experiences of nalbuphine hydrochloride. According to references, combinations of nalbuphine and opioids decrease incidence of opioid related side effects, ex: pruritus.

When NALDEBAIN combine used with nalbuphine, the dose of nalbuphine should not exceed 80 mg per day or 20 mg Q6H.

General anesthetic

Studies of NALDEBAIN concomitant use with general anesthetic, including inhaled anesthetic, intravenous administered anesthetic like opioid and benzodiazepine, have not been conducted. Since NALDEBAIN is nalbuphine's prodrug, the concomitant use with general anesthetics could refer to experiences of nalbuphine hydrochloride. According to references, there is no clinical significant safety concern without dosage adjustment of anesthetic.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid- related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue Naldebainif serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

8 USE IN SPECIFIC POPULATIONS

Pregnancy

Teratogenic Effects: Pregnancy Category B

Reproduction studies have been performed in rats by subcutaneous administration of nalbuphine up to 100 mg/kg/day, or 590 mg/m2/day which is approximately 6 times the MRHD(Maximum Recommended Human Dose), and in rabbits by intravenous administration of nalbuphine up to 32 mg/kg/day, or 378 mg/m2/day which is approximately 4 times the MRHD. The results did not reveal evidence of developmental toxicity, including teratogenicity, or harm to the fetus. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, NALDEBAIN should be used during pregnancy only if clearly needed.

Non-teratogenic Effects:

Neonatal body weight and survival rates were reduced at birth and during lactation when nalbuphine was subcutaneously administered to female and male rats prior to mating and throughout gestation and lactation or to pregnant rats during the last third of gestation and throughout lactation at doses approximately 4 times the maximum recommended human dose.

Labor and Delivery

See 5.4.

Nursing Mothers

Limited data suggest that nalbuphine (nalbuphine hydrochloride) is excreted in maternal milk but only in a small amount (less than 1% of the administered dose) and with a clinically insignificant effect. Caution should be exercised when NALDEBAIN is administered to a nursing woman.

Pediatric Use

Safety and effectiveness in pediatric patients have not been established.

Geriatric Use

Not necessary to adjust dose in geriatric population.

9 OVERDOSAGE

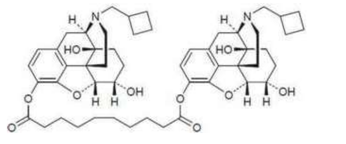
There is no incidence of NALDEBAIN administering overdose in clinical trial. The drug is supplied in 2-mL vial for single dose usage with no risk of overdosage.

The suggestion for overdosage is immediate intravenous administration an opiate antagonist such as naloxone or nalmefene is a specific antidote. Oxygen, intravenous fluids, vasopressors and other supportive measures should be used as indicated.

The administration of single doses of 72 mg of nalbuphine subcutaneously to eight normal subjects has been reported to have resulted primarily in symptoms of sleepiness and mild dysphoria.

10 DESCRIPTION

NALDEBAIN Extended Release Injection, a prodrug of nalbuphine, contains dinalbuphine sebacate as the active ingredient.



NALDEBAIN Extended Release Injection is a sterile, clear and light yellow oily solution. The product is packed in a 2-mL vial used for muscular injection. The product is supplied in a 2-mL glass vial.

Inactive ingredients include, benzyl benzoate, and sesame oil.

11 CLINICAL PHARMACOLOGY

Mechanism of Action

Dinalbuphine sebacate is a synthetic nalbuphine prodrug. The dosage form is a sterile oil solution, which is suitable for intramuscular injection. Dinalbuphine sebacate contains two nalbuphine molecules joined by sebacoyl ester which is rapidly hydrolyzed to nalbuphine by esterase. Nalbuphine is the active moiety.

Nalbuphineis a potent analgesic. Its analgesic potency is essentially equivalent to that of morphineon a milligram basis. Receptor studies show that nalbuphinebinds to mu, kappa, and delta receptors, but not to sigma receptors. Nalbuphineis primarily a kappaagonist/partial mu antagonist analgesic.

Pharmacodynamics

The pharmacodynamics of dinalbuphine sebacate is from nalbuphine.

Nalbuphine may produce the same degree of respiratory depression as equianalgesic doses of morphine. However, nalbuphine exhibits a ceiling effect such that increases in dose greater than 30 mg do not produce further respiratory depression in the absence of other CNS active medications affecting respiration.

Nalbuphine by itself has potent opioid antagonist activity at doses equal to or lower than its analgesic dose. When administered following or concurrent with mu agonist opioid analgesics (e.g., morphine, oxymorphone, fentanyl), nalbuphine may partially reverse or block opioid-induced respiratory depression from the mu agonist analgesic. Nalbuphine may precipitate withdrawal in patients dependent on opioid drugs. NALDEBAIN should be used with caution in patients who have been receiving mu opioid analgesics on a regular basis.

Pharmacokinetics

Absorption

Following intramuscular administration 150mg of dinalbuphine sebacate, the drug is absorbed and rapidly hydrolyzed as nalbuphine with peak concentrations (C_{max}) achieved at 64.0±9.3 hours. The mean C_{max} is estimated to be 15.4±6.4 ng/mL.

Metabolism

Dinalbuphine sebacate is metabolized primarily by esterase. Biotransformation studies showed that over 90% of the prodrug was converted to nalbuphine in about 30 min in fresh human whole blood.

Nalbuphine is metabolized by Cytochrome P450s and phase II enzyme UGTs (uridinyl diphosphate glucuronosyltransferases) and produce glucuronide metabolites.

Distribution

The mean apparent volume of distribution in healthy volunteers after administration 150 mg NALDEBAIN is estimated to be 10628 ± 4403 L

In vitro plasma protein binding study suggest that dinalbuphine sebacate protein binding is about 90% in human plasma.

Dinalbuphine sebacate and nalbuphine did partition into red blood cells but not to a greater extent than to plasma. The RBC partition coefficients, $K_{RBC/PL}$, for dinalbuphine sebacate determined to be 1.20 ; nalbuphine determined to be 1.24.

Excretion

Nalbuphine is mainly excreted by the kidneys. Following intramuscular administration 150mg of dinalbuphine sebacate, the drug is absorbed and rapidly hydrolyzed as nalbuphine. The elimination half-life of nalbuphine was 83.2±46.4 hr. Mean clearance of nalbuphine is 100±11 L/h. Less than 4% nalbuphine of each dose was recovered in urine.

Drug Interaction

No drug interaction studies have been conducted [see Drug Interactions (7)].

12 NONCLINICAL TOXICOLOGY

Carcinogenesis, Mutagenesis, Impairment of Fertility

Carcinogenesis

No carcinogenesis study was conducted with dinalbuphine sebacate. According to reference, long term carcinogenicity studies were performed in rats (24 months) and mice(19 months) byoral administration at dosesup to 200 mg/kg (1180 mg/m²) and 200 mg/kg (600 mg/m²) perday, respectively. Therewasnoevidenceofan increase in tumors in either species related to nalbuphine administration.

Mutagenesis

Dinalbuphine sebacate did not show genotoxic activity in the in vivo mouse peripheral blood micronucleus assay.

According to reference, nalbuphine did not have mutagenic activity in the AMES test with four bacterial strains, in the Chinese Hamster Ovary HGPRT assays or in the Sister Chromatids Exchange Assay. However, nalbuphine induced an increased frequency of mutation in the mouse lymphoma assay. Clastogenic activity was not observed in the mouse micronucleus test of the cytogenicity bone marrow assay in rats.

Impairment of Fertility

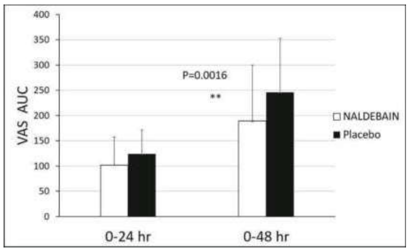
No reproduction toxicity study was conducted with dinalbuphine sebacate. In reproductive and developmental toxicity studies in rats, nalbuphine did not affect fertility at subcutaneous doses up to 56 mg/kg/day or 330 mg/m²/day.

13 CLINICAL STUDIES

A multicenter, randomized, double-blind, placebo-controlled phase III study evaluated the safety and efficacy of 150 mg NALDEBAIN in patients undergoing hemorrhoidectomy. There are 78 subjects in NALDEBAIN group while 78 subjects in placebo group. 24±12hrs prior to hemorrhoidectomies, the subjects were treated with single IM injection of NALDEBAIN 150mg. The post-operative analgesics were PCA dosing with ketorolac as needed for the first two days (Day 1 and 2) and oral dosing with ketorolac as needed for the following 5-8 days (Day 3 to 10). The primary objective of this study is pain assessment (time-specific pain intensity) calculated as the area under the curve (AUC) of VAS pain intensity scores through 48 hours after surgery ($AUC_{0\text{-}48}$). There was a significant treatment effect for NALDEBAIN compared to placebo($p=0.0016$) (Figure 1).

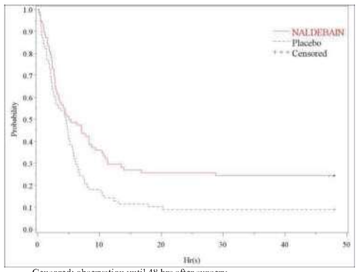
In this clinical study, NALDEBAIN demonstrated a significant reduction in pain intensity compared to placebo for up to 48 hours after surgery, equals to 72 hours after administration. The difference in mean pain intensity between treatment groups occurred during the 24 and 48 hours after surgery, which means following 48 and 72 hours study drug administration. Subjects treated with NALDEBAIN took significantly longer periods of time for the first use of Ketorolac via PCA than those in placebo group, 10.90 hours in NALDEBAIN group, 5.90 hours in placebo group. Figure 2 shows the probability of subjects who does not use the PCA within 48 hours after surgery. The total amount of Ketorolac administered by PCA through 48 hours after surgery in NALDEBAIN group is 43.04 mg while in placebo group is 77.81 mg. From Day 3 to Day 7 after surgery, there was a significant decrease in oral analgesics consumption, 47.95 mg in NALDEBAIN group, 67.05 mg in placebo group. The results for assessments of pain intensity measured with VAS scores during Day 3-7 listed in table 3. Besides, in post-hoc analysis, pain assessment during Day 3-7 was calibrated for the use of rescue medication as that was done for the primary end point. After calibration, the pain intensity (VAS scores) in NALDEBAIN group is less than placebo group. In post-hoc analysis, there was difference between NALDEBAIN and placebo treatments on VAS $AUC_{0\text{-}final}$ (Figure 3).

Figure 1. Cumulative pain score (VAS AUC) through 24 hours and 48 hours after hemorrhoid operation.-mITT



** p<0.01 AUC: area under the curve.

Figure 2 Kaplan-Meier graph of time from post-operation to the first use of PCA Ketorolac (the probability of the events not happening within a time period)- mITT



Censored: observation until 48 hrs after surgery

Table 3 Statistical analysis of VAS scores for pain intensity during Day 3-7 by day and treatment-mITT

Day	Mean ± SD (NObs[1]) NALDEBAIN	Mean ± SD (NObs[1]) Placebo	SDE – Placebo LS-mean [95% CI][2]
Day 3			
Morning	2.65 ± 2.30 (77)	2.46 ±2.21 (78)	0.234 [-0.46; 0.93]
Evening	3.47 ± 2.83 (77)	3.87 ± 2.80 (75)	- 0.268 [-1.84; 1.31]
Day 4			
Morning	3.39 ± 2.60 (78)	3.68 ± 2.82 (78)	-0.278 [-1.14; 0.59]
Evening	3.20 ± 2.81 (76)	3.50 ± 2.70 (77)	-0.134 [-1.01; 0.75]
Day 5			
Morning	3.10 ± 2.66 (77)	3.33 ± 2.47 (77)	-0.206 [-1.01; 0.60]
Evening	3.29 ± 2.86 (74)	3.21 ± 2.64 (77)	-0.599 [-2.10; 0.90]
Day 6			
Morning	3.04 ± 2.67 (74)	3.03 ± 2.57 (77)	0.030 [-0.79; 0.85]
Evening	2.65 ± 2.45 (75)	2.80 ± 2.57 (75)	-0.121 [-0.92; 0.68]
Day 7			
Morning	2.50 ± 2.40 (74)	2.93 ± 2.51 (74)	-0.438 [-1.21; 0.33]

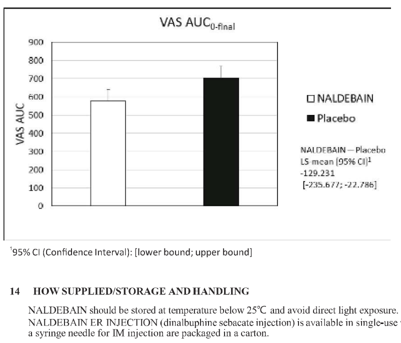
[1]NObs = Number of Observation
[2]95% CI (Confidence Interval): [lower bound; upper bound]

Table 4 Statistical analysis of adjusted VAS scores for pain intensity during Day 3-7 by day and treatment–mITT

Day	Mean ± SD (NObs[1]) NALDEBAIN	Mean ± SD (NObs[1]) Placebo	SDE – Placebo LS-mean [95%
Day 3			
Morning	3.565 ± 2.787 (78)	4.977 ± 3.197 (78)	-1.395 [-2.339; -0.451]
Evening	4.286 ± 3.027 (78)	5.406 ± 3.107 (78)	-1.105 [-2.083; -0.127]
Day 4			
Morning	4.104 ± 2.937 (78)	4.708 ± 3.281 (78)	-0.627 [-1.618; 0.364]
Evening	4.394 ± 3.216 (77)	5.225 ± 3.373 (77)	-0.832 [-1.891; 0.226]
Day 5			
Morning	4.136 ± 3.044 (77)	4.767 ± 3.228 (77)	-0.643 [-1.652; 0.366]
Evening	4.355 ± 3.286 (75)	5.244 ± 3.191 (77)	-0.413 [-2.251; 1.425]
Day 6			
Morning	3.831 ± 2.974 (75)	4.344 ± 3.330 (77)	-0.509 [-1.526; 0.509]
Evening	3.507 ± 3.062 (75)	4.376 ± 3.489 (75)	-0.845 [-1.903; 0.212]
Day 7	3.007 ± 2.808 (75)	4.259 ± 3.210 (75)	-2.422 [-4.100; -0.743]

[1]NObs = Number of Observation
[2]95% CI (Confidence Interval): [lower bound; upper bound]

Figure 3 Statistical analysis of $AUC_{0\text{-}final}$ by treatment after hemorrhoid operation-mITT



[1]95% CI (Confidence Interval): [lower bound; upper bound]

14 HOW SUPPLIED/STORAGE AND HANDLING

NALDEBAIN should be stored at temperature below 25℃ and avoid direct light exposure. Please store in the carton before usage. NALDEBAIN ER INJECTION (dinalbuphine sebacate injection) is available in single-use vials. 2 mL single use vial (75 mg/mL) and a syringe needle for IM injection are packaged in a carton.

Product owner:

Lumosa Therapeutics Co., Ltd.

4F, No. 3-2, Park Street, Nangang District, Taipei, 11503, Taiwan

Manufacturer:

Hsinchu Plant of UBI Pharma Inc.

No.45, Guangfu N. Rd., Hukou Township, Hsinchu County 303036 , Taiwan, R.O.C. Version:

MY_V1.2

Reg number: MAL22066003ACRZ
Revision date: Jun 2022

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